COMP7610 Smart Contract Lab – Assignment 3

HKU Drone Club: CPT and Campus-Memo

Student: [REPLACE: Full name] UID: [REPLACE: UID] Network: Sepolia (Chain ID 11155111)

System design. `CPT` is an OpenZeppelin ERC-20 token with 18 decimals. Deployment mints exactly 1,000 CPT to the CPT contract as the club treasury. Only `ADMIN_ROLE` can call `mintPoints` (new points) or `distributePoints` (treasury points). Standard `transfer`, `approve`, and `transferFrom` remain available to users.

Members call `applyForMembership`; an administrator calls `approveMembership`, which grants membership and transfers a fixed 50 CPT welcome allocation from the treasury. A member can call `withdrawFromClub` to burn all remaining CPT and revoke membership. Existing NFTs are not destroyed. This is the additional custom withdrawal function.

`CampusMemo` is an ERC-721 collection named "Campus-Memo". Its maximum supply is 10 and minting costs 100 CPT. A member first approves the NFT contract for 100 CPT, then calls `mintMemo`; the contract calls CPT `burnFrom`, so the payment is removed from total supply. Token IDs start at 1 and `tokenURI(id)` is the IPFS base URI plus the token ID.

Local verification.

```
npm install
npm run compile
npm test
```

The test suite checks the initial treasury supply, administrator permissions, welcome points, CPT transfer, withdrawal burn, preservation of an already-owned NFT after withdrawal, NFT payment burn, NFT transfer, URI generation, and rejection of the 11th NFT.

Sepolia deployment (Hardhat).

1. In MetaMask select Sepolia and obtain test ETH. Never use a mainnet private key.
2. Set environment variables locally (do not commit them):

```
export SEPOLIA_RPC_URL="<RPC URL>"
export DEPLOYER_PRIVATE_KEY="<test-wallet private key>"
export NFT_BASE_URI="ipfs://<CID>/"
```

3. Run `npm run compile`, then `npm test`, then `npm run deploy:sepolia`.
4. Record the printed addresses and deployment transaction hashes. Verify both addresses on Sepolia Etherscan.

Remix + MetaMask function test.

1. Open the two Solidity files in Remix, compile with Solidity 0.8.24, choose "Injected Provider – MetaMask", and confirm Sepolia.
2. Deploy `CPT` with the deployer's address. Deploy `CampusMemo` with the CPT address and `ipfs://<CID>/`.
3. As a student account call `applyForMembership`; as the administrator call `approveMembership(student)`. Confirm `isMember=true` and a 50 CPT balance.
4. As the administrator call `distributePoints(student, 100 CPT)`. The student calls `approve(CampusMemo, 100 CPT)`, then `mintMemo()`. Confirm `ownerOf(1)` and `tokenURI(1)`.
5. From the student account call `transfer` to another account and `safeTransferFrom` for token 1. Confirm both new balances/owners.
6. From the member account call `withdrawFromClub()`. Confirm the CPT balance is zero, `isMember=false`, the `MemberWithdrawn` event is emitted, and `ownerOf(1)` is still the former member.

Final submission values.

CPT address	[REPLACE: paste deployed Sepolia address]
Campus-Memo address	[REPLACE: paste deployed Sepolia address]
NFT base URI	[REPLACE: paste IPFS base URI]
Example token URI	[REPLACE: paste tokenURI(1) result]
Deployment transactions	[REPLACE: paste transaction hashes or Etherscan links]

Screenshots to insert before submission: (1) MetaMask showing Sepolia and the deployed contracts; (2) membership approval and 50 CPT balance; (3) CPT approval followed by NFT mint, `ownerOf(1)`, and `tokenURI(1)`; (4) withdrawal transaction, zero CPT balance, `isMember=false`, and unchanged `ownerOf(1)`. Rename this file to `FullName_UID_Assignment3_Report.tex` and export the same content as PDF after replacing every red placeholder.